$$A = \left\{ \frac{3n-1}{n+2} : n \in \mathbb{N} \right\}$$

1) MOSTRAR QUE 3 ES COTA SUPERIOR.
QUEREMOS VER QUE

$$\frac{3n-1}{n+2} \leq 3 \quad , \forall n \in \mathbb{N}$$

$$\Updownarrow$$

$$3n-1 \leq 3.(n+2)$$

$$\Downarrow$$

$$3n-1 \leq 3n+6$$

$$\Downarrow$$

$$-1 \leq 6 \rightarrow \underline{\text{VERDADERO } \forall n \in \mathbb{N}}.$$

$$\begin{array}{r|l} 3n-1 & n+2 \\ -3n+6 & 3 \\ \hline -7 & \end{array} \qquad \Rightarrow 3n-1 = 3(n+2) - 7$$

$$\frac{3n-1}{n+2} = \frac{3(n+2)-7}{n+2} =$$

$$= \frac{3\cancel{(n+2)}^{1}}{\cancel{n+2}^{1}} - \frac{7}{n+2} = 3 - \frac{7}{n+2} \leq 3 \quad \checkmark$$

OBS $\quad \frac{3n-1}{n+2} < 3 \quad , \forall n \in \mathbb{N}.$

$\therefore \underline{3 \notin A}$.

2) HALLAR UN $a \in A$ / $a > 2{,}99$.
BUSCAMOS $n \in \mathbb{N}$ TAL QUE

$$\frac{3n-1}{n+2} > 2{,}99$$

$$3 - \frac{7}{n+2} > 2,99$$

$$-\frac{7}{n+2} > 2,99 - 3$$

$$-\frac{7}{n+2} > -0,01$$

$$\frac{\not 7}{\not 0,01} < n+2$$

$$700 < n+2$$

$$\underline{698 < n}$$

∴ TOMANDO $n = 699$, OBTENEMOS

$$3 - \frac{7}{699+2} > 2,99$$

$$\cong 2,990014$$

$$\in A$$

3) MOSTRAR QUE SI $t < 3$, ENTONCES EXISTE $a \in A$ TAL QUE TAL QUE $a > t$.

SEA $t < 2$.

$$3 - \frac{7}{n+2} > t$$

$$-\frac{7}{n+2} > t - 3$$

$$\frac{1}{n+2} < \frac{t-3}{-7} = \frac{3-t}{7}$$

$$\frac{7}{3-t} < n+2$$

$$n+2 > \frac{7}{3-t}$$

$$n > \frac{7}{3-t} - 2$$

$\frac{7}{3-t} > 0$

$\therefore$ PARA $t<3$, BASTA TOMAR $n > \frac{7}{3-t} - 2$.

POR EJEMPLO SI TOMAMOS

$$n = \max\left\{ \left[\frac{7}{3-t} - 2 \right], 1 \right\},$$

$\left[\frac{7}{3-t} - 2 \right]$ → PARTE ENTERA



ENTONCES $a = \frac{3n-1}{n+2} = 3 - \frac{7}{n+2} > t$

OBS: POR 1) Y 3), RESULTA QUE

$3 = \sup(A)$.

OBS: $3 \underline{\text{NO}}$ ES $\max(A)$ PORQUE

$3 \notin A$.

- SI $s = \sup(A) \in A$, ENTONCES

$s = \max(A)$ (MÁXIMO)

- Si $i = \inf(A) \in A$, entonces

$$i = \min(A).$$

Ej

